



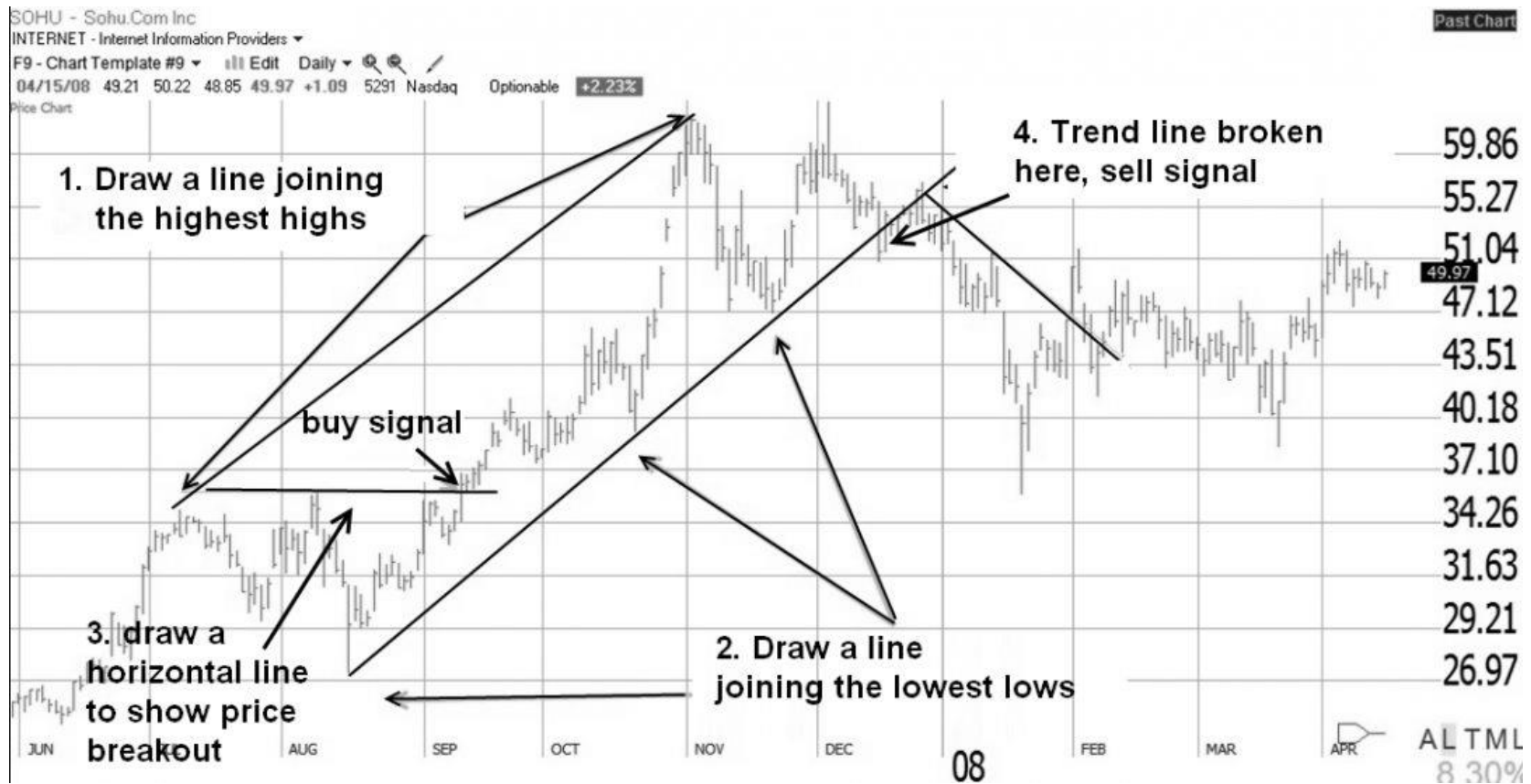
Graphical Perception

ONE LOVE. ONE FUTURE.

What is graphical perception?

- The human capacity for visually interpreting information on graphs and charts
- The visual decoding of information encoded on graphs
 - It is a complex process that involves multiple cognitive abilities, including **attention**, **pattern recognition**, and **estimation**.

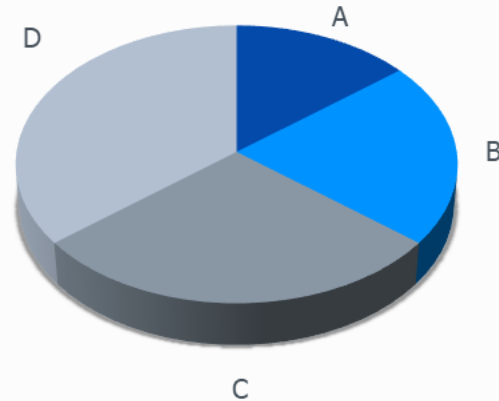
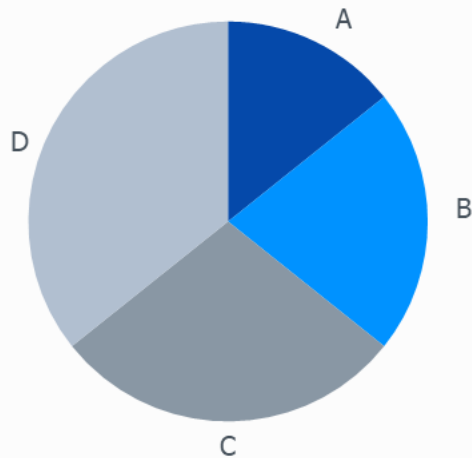
Attention, pattern recognition, estimation



Why study graphical perception

- Unlocking effective communication
 - Ensures clear and impactful communication of insights.
- Enhancing interpretation speed
 - Graphical perception taps into the brain's visual processing capabilities.
 - Mastery accelerates interpretation, enabling quick extraction of meaningful insights from visualized data.
- Reducing cognitive load
 - Optimizing visual elements minimizes cognitive strain on viewers.
 - Graphical perception helps design visualizations that align with the brain's natural processing, reducing the mental effort required for information absorption.

Good vs. bad visualization



Why important?

“Graphical excellence is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space” — Edward Tufte



Topics

- Signal Detection
- Magnitude Estimation
- Pre-Attentive Processing
- Using Multiple Visual Encodings
- Gestalt Grouping
- Change Blindness

Signal Detection

Detecting brightness



A

Which is brighter?



B

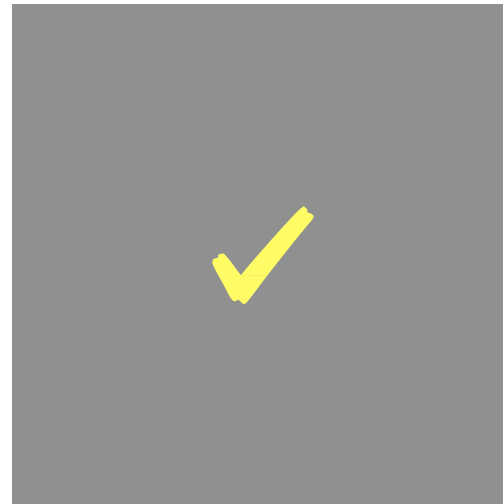
Detecting Brightness

(128,128,128)



A

(144,144,144)



B

Detecting Brightness



A

Which is brighter?



B

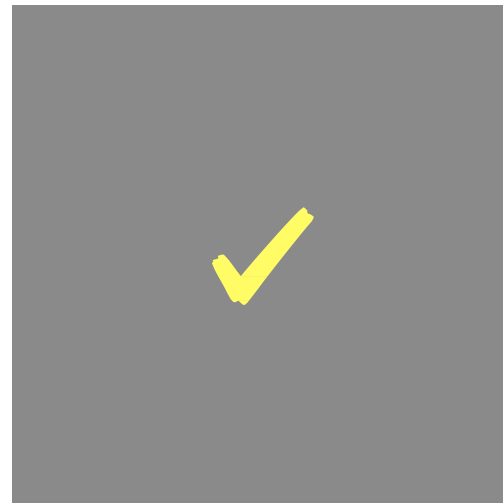
Detecting Brightness

(134,134,134)



A

(138,138,138)



B

Weber's Law: Just Noticeable Difference (JND)

- also known as the difference threshold, is the minimum level of stimulation that a person can detect 50 percent of the time.

$$dp = k \frac{dS}{S}$$

Weber's Law: Just Noticeable Difference (JND)

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$$dp = k \frac{dS}{S}$$

← Change of Intensity

← Physical Intensity

Weber's Law: Just Noticeable Difference (JND)

- also known as the difference threshold, is the minimum level of stimulation that a person can detect 50 percent of the time.

Perceived Change $\longrightarrow$ $dp = k \frac{dS}{S}$ $\longleftarrow$ Change of Intensity

$\longleftarrow$ Physical Intensity

Weber's Law: Just Noticeable Difference (JND)

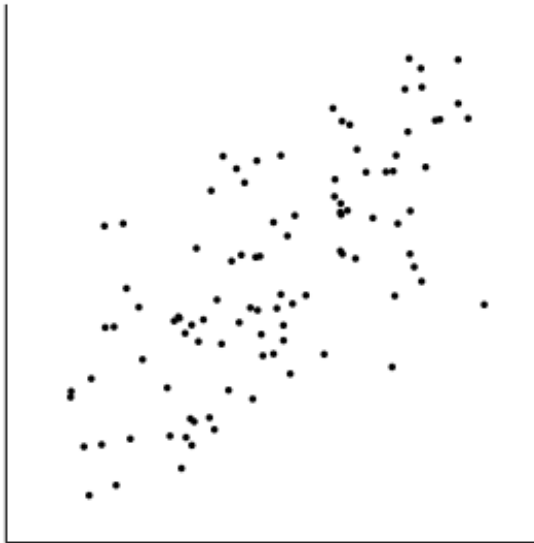
Perceived Change $\longrightarrow$ $dp = k \frac{dS}{S}$ $\longleftarrow$ Change of Intensity
 $\longleftarrow$ Physical Intensity

Most continuous variation in stimuli are perceived in discrete steps

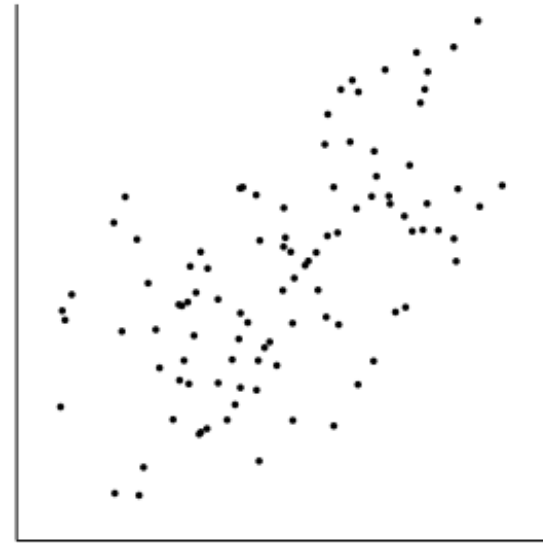


Ranking correlation visualizations

Which of the two appeared to be more highly correlated?



A



B

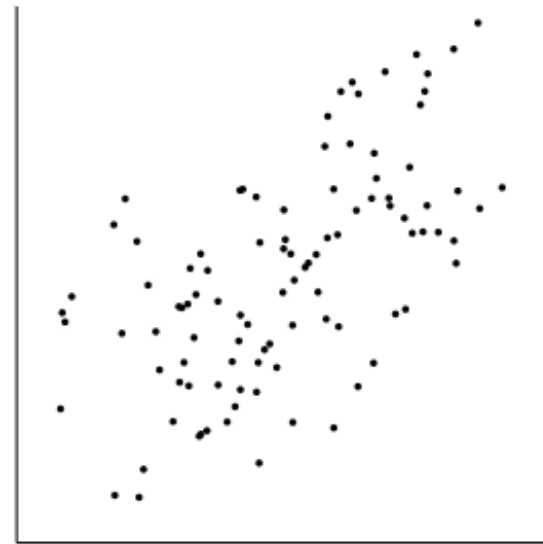
[Harrison et al 2014]

Ranking correlation visualizations

Which of the two appeared to be more highly correlated?



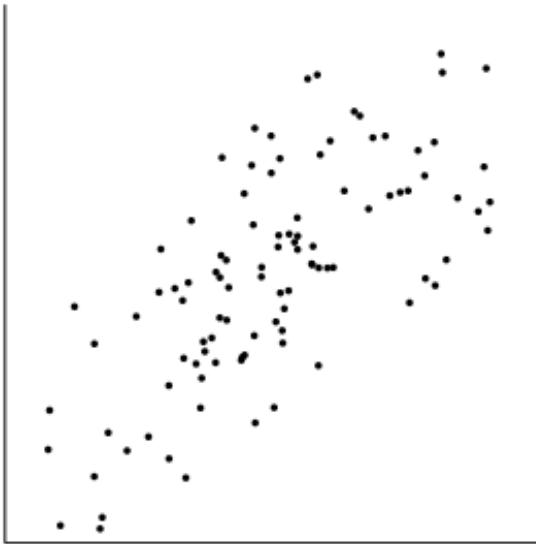
$r = 0.7$



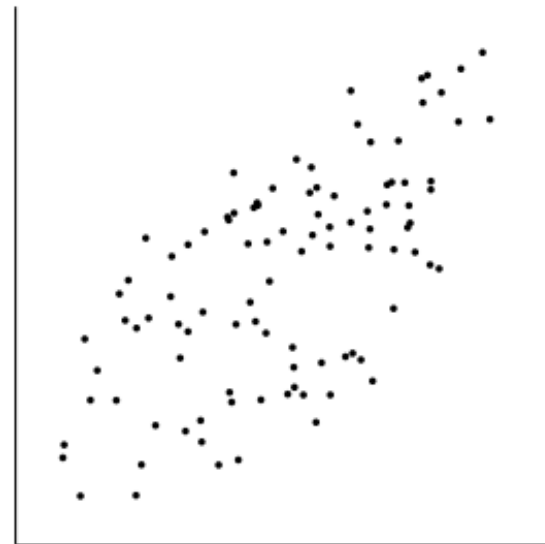
$r = 0.6$

Ranking correlation visualizations

Which of the two appeared to be more highly correlated?



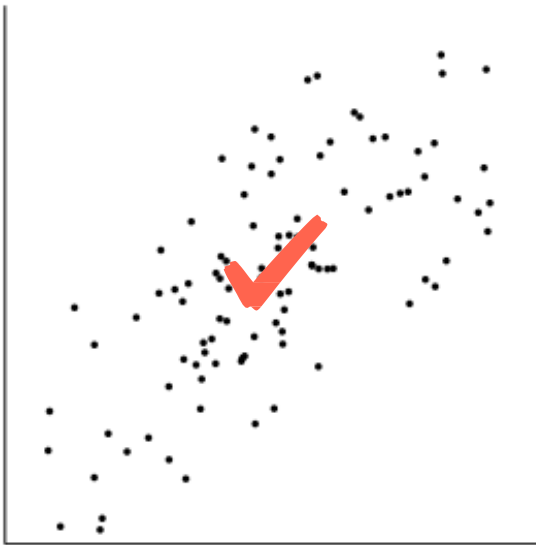
A



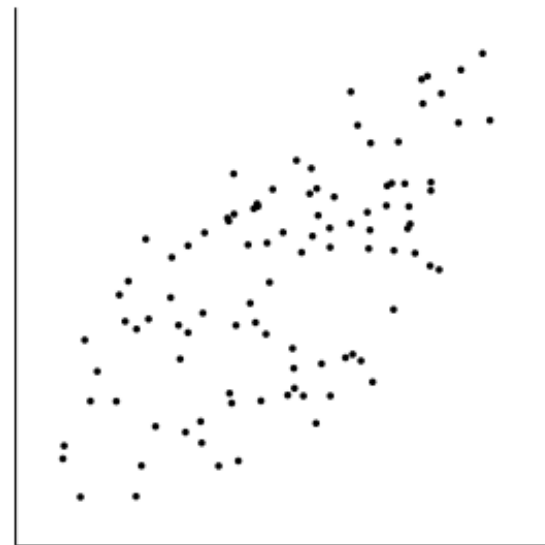
B

Ranking correlation visualizations

Which of the two appeared to be more highly correlated?



$r = 0.7$



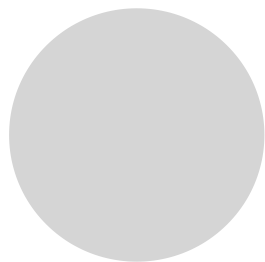
$r = 0.65$

Magnitude Estimation

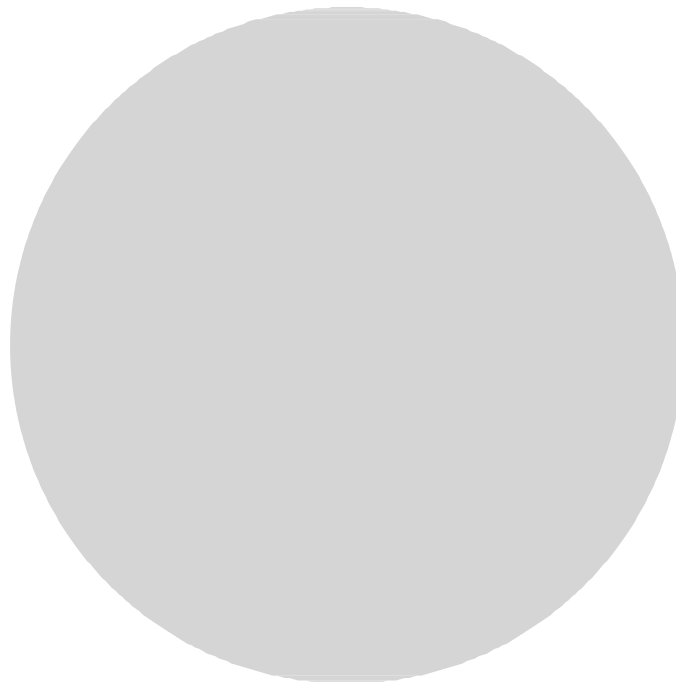
A Quick Experiment...



A



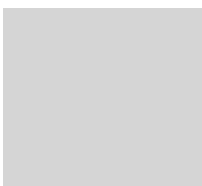
B



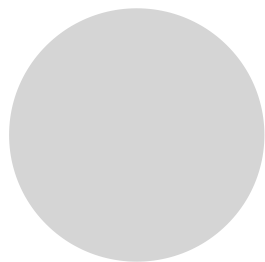
B



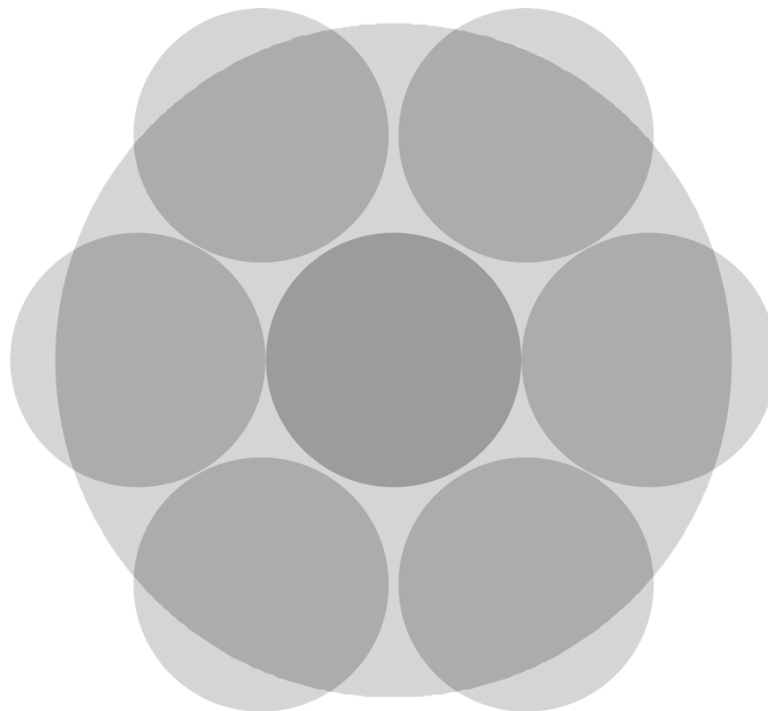
A



A



B

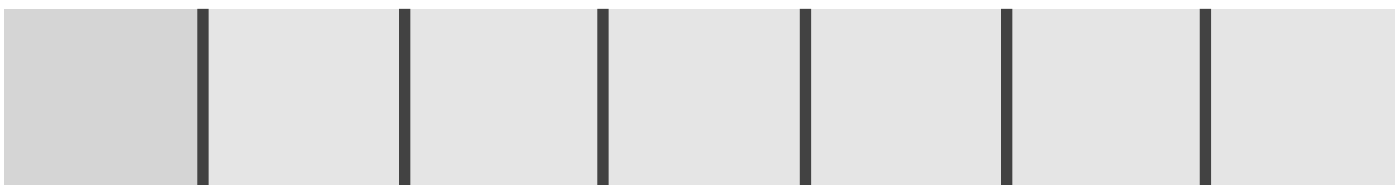


Area

B



A



Length

Steven's Power Law

[Graph from T. Munzner 2014]

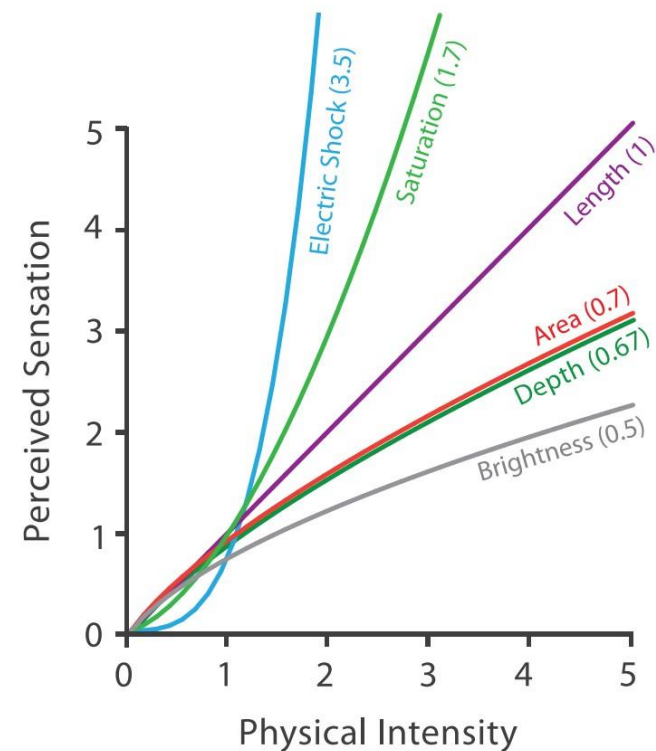
Models the **relationship** between the **magnitude** of a physical stimulus and its perceived intensity.

Exponent
(Empirically Determined)

$$S = I^p$$

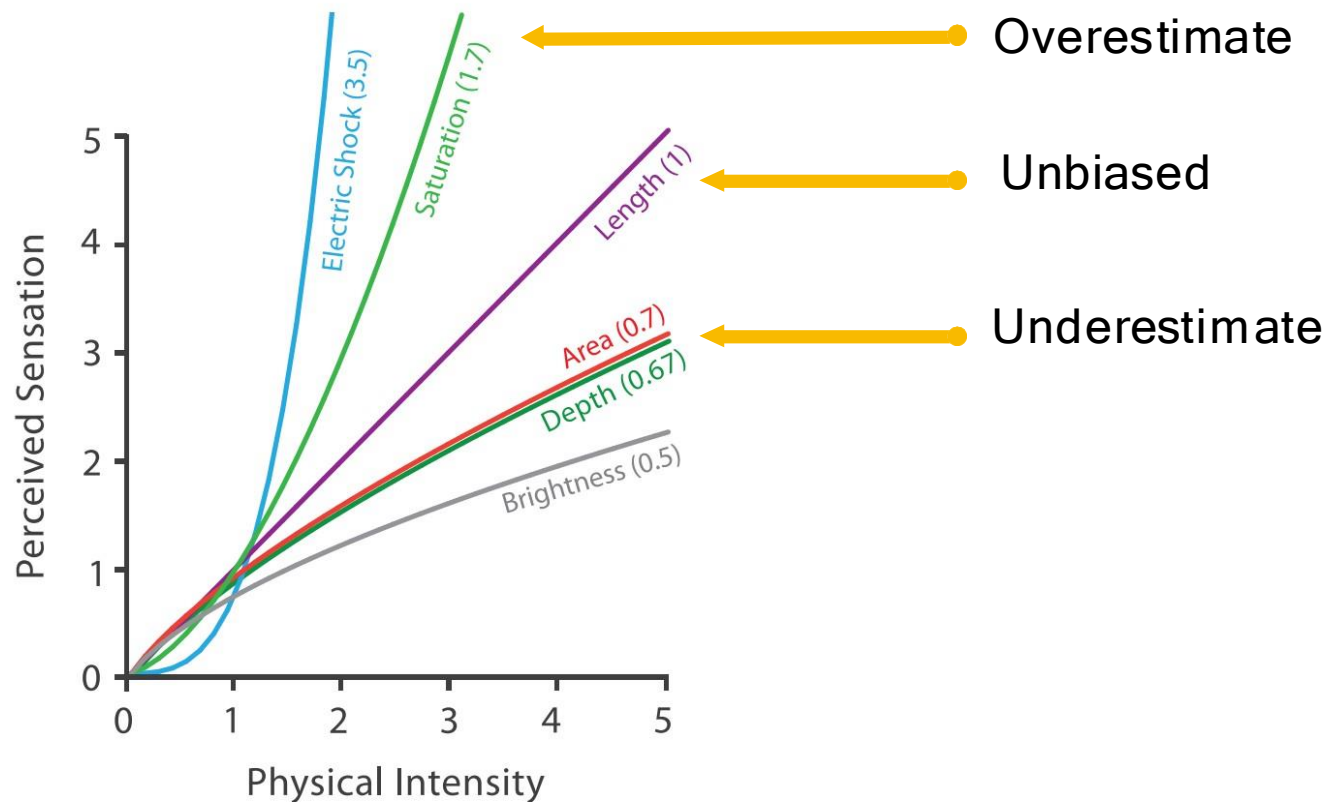
Perceived Sensation Physical Intensity

Predicts bias, not necessarily accuracy!



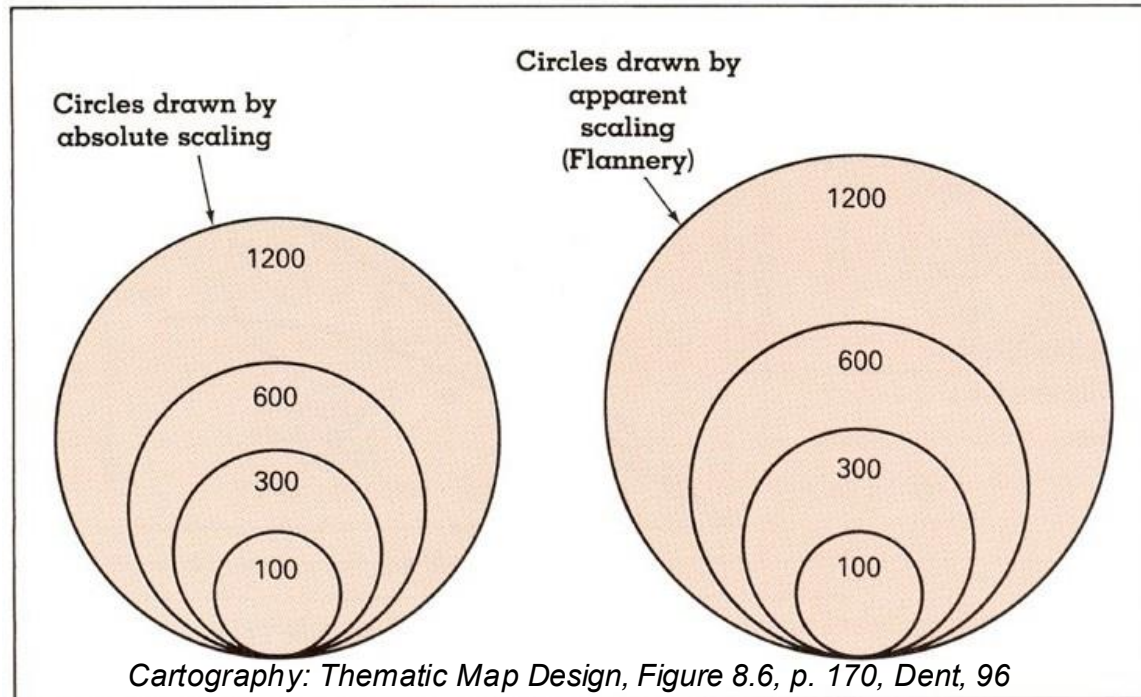
Steven's Power Law

[Graph from T. Munzner 2014]



Apparent Magnitude Scaling

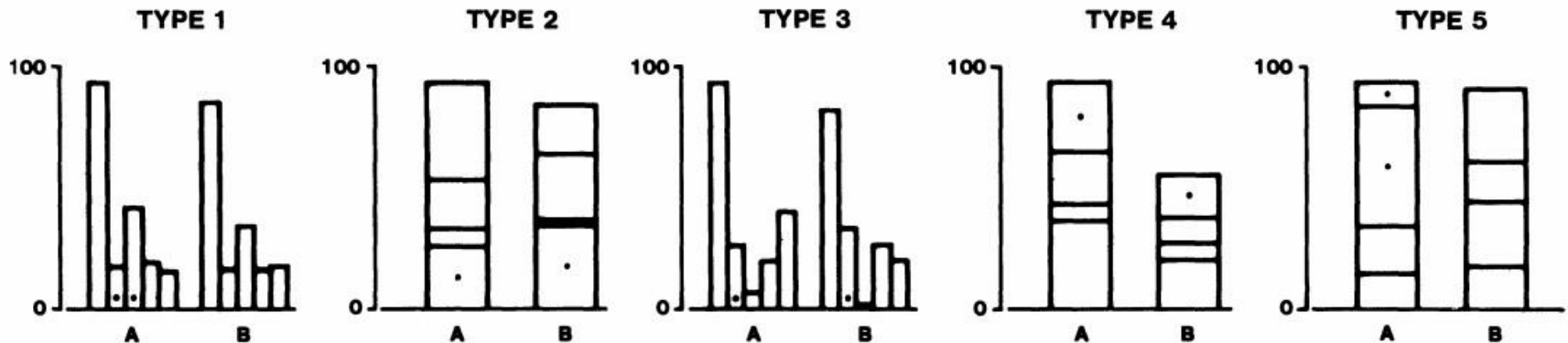
- To compensate for human error in interpreting scale because people tend to underestimate area



$$\times \frac{1}{0.7}$$

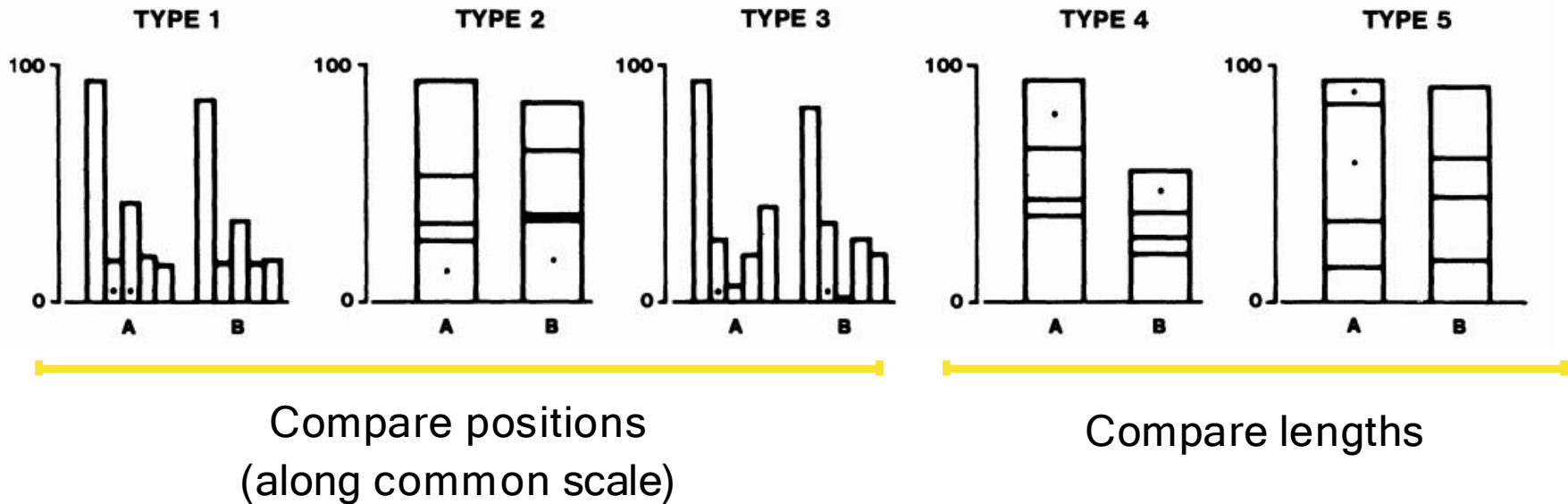
Graphical Perception [Cleveland & McGill 84]

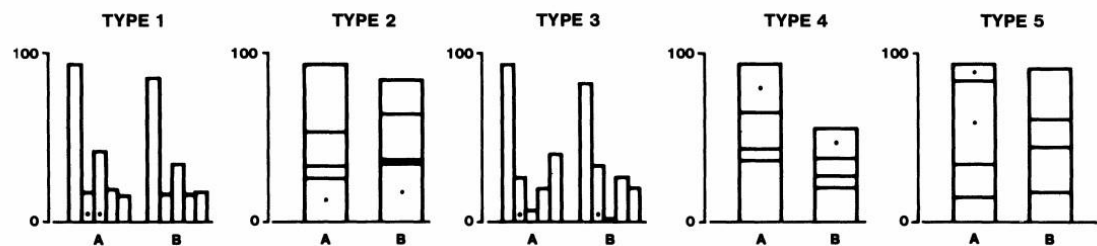
- What percentage of the smaller was of the larger?



Graphical Perception [Cleveland & McGill 84]

- What percentage of the smaller was of the larger?





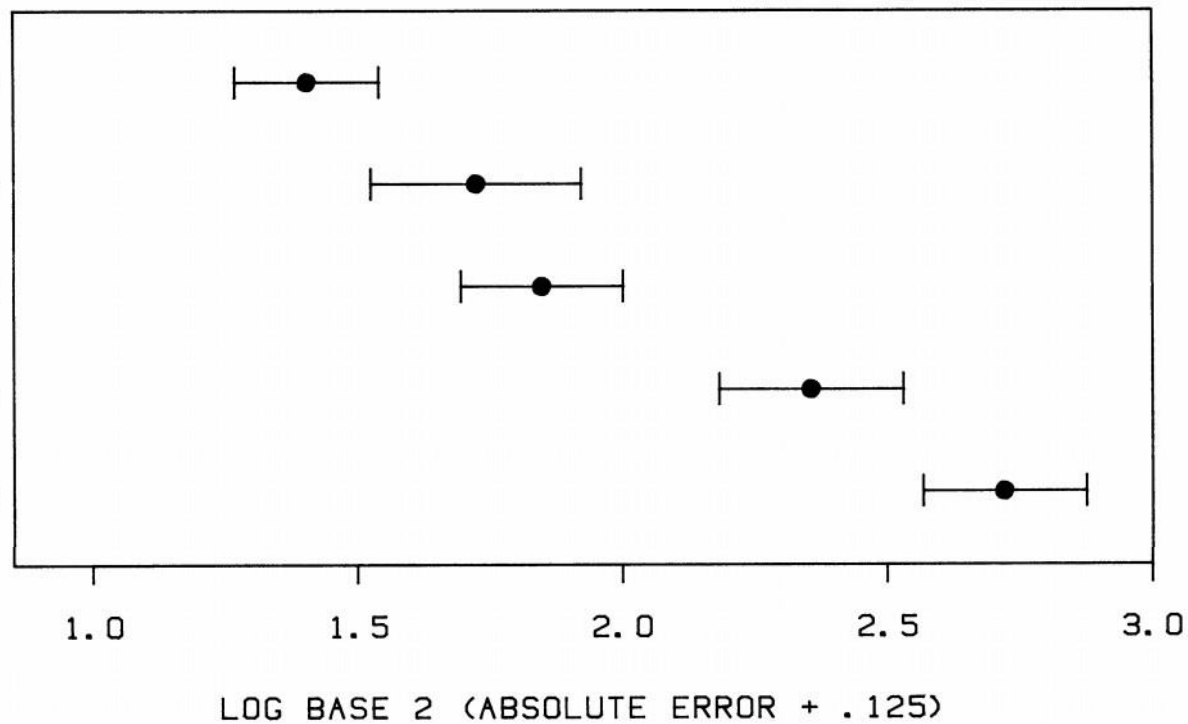
TYPE 1 (POSITION)

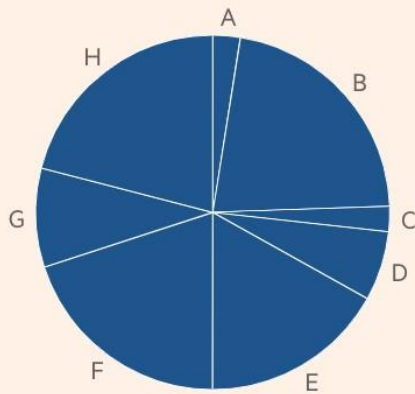
TYPE 2 (POSITION)

TYPE 3 (POSITION)

TYPE 4 (LENGTH)

TYPE 5 (LENGTH)





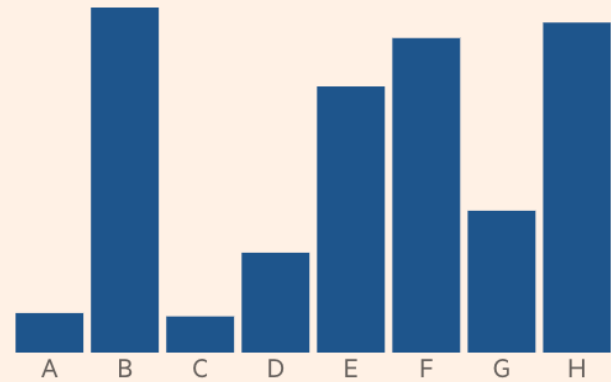
Which is the third largest segment in the pie chart?

E

F

G

H



Which is the 3rd largest bar?

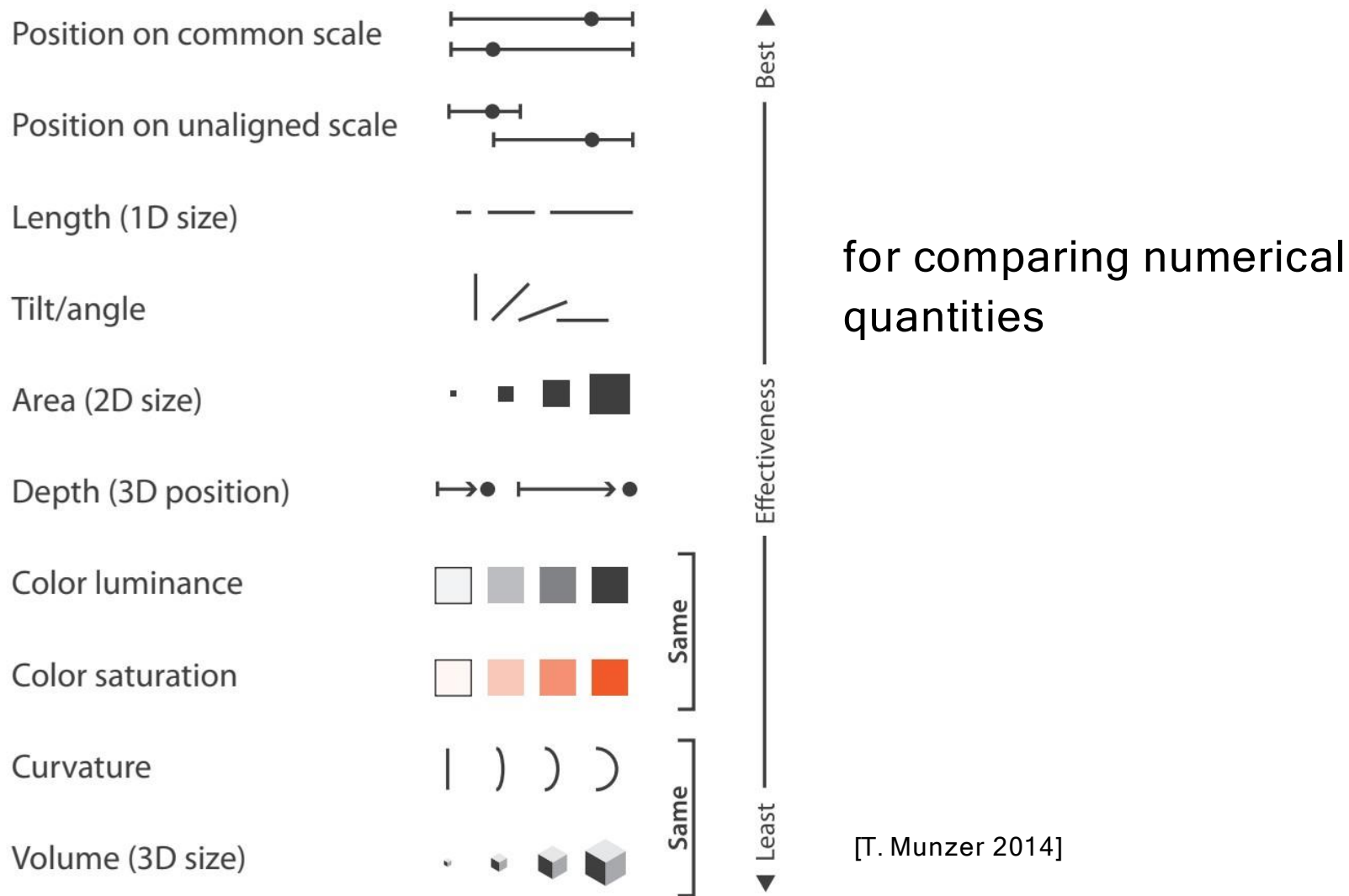
E

F

G

H

Effectiveness Ranking of Visual Encoding Variables



Pre-Attentive Processing

How Many 3's?

1281768756138976546984506985604982826762
9809858458224509856458945098450980943585
9091030209905959595772564675050678904567
8845789809821677654876364908560912949686

[based on a slide from J Stasko]

How Many 3's?

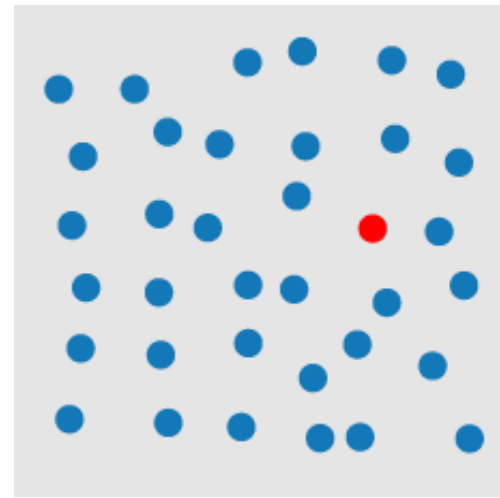
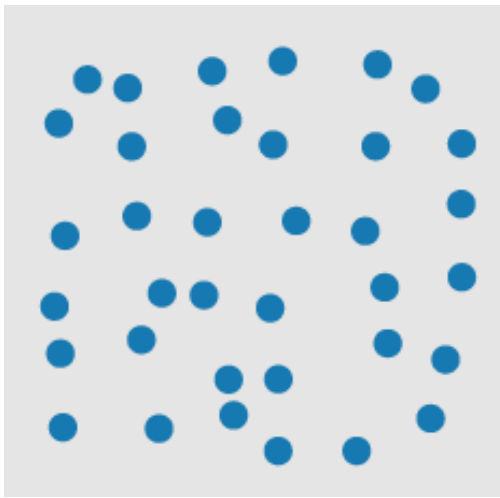
12817687561**3**8976546984506985604982826762
980985845822450985645894509845098094**3**585
90910**3**0209905959595772564675050678904567
8845789809821677654876**3**64908560912949686

Pre-attentive processing

- The ability of the low-level human visual system to effortlessly identify certain basic visual properties.

Visual Pop-Out: Color

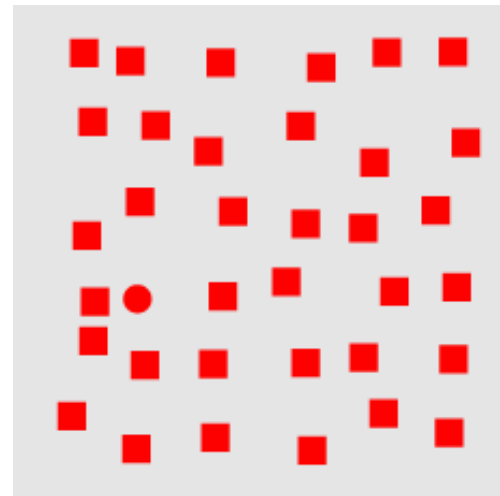
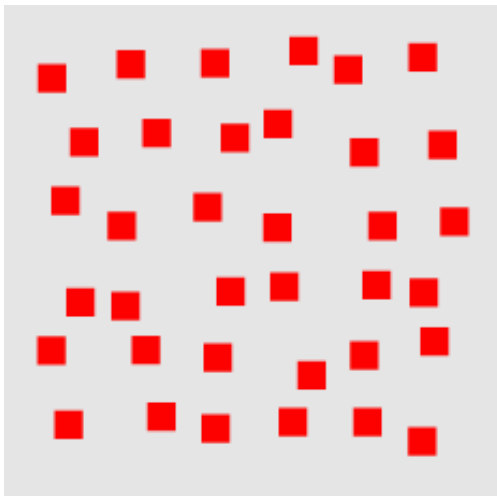
- A unique visual property in the target allows it to "pop out" of a display



www.csc.ncsu.edu/faculty/healey/PP/index.html

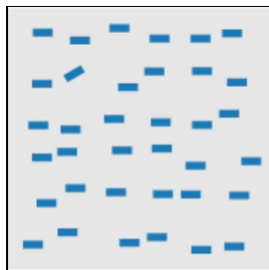
Visual Pop-Out: Shape

- A unique visual property in the target allows it to "pop out" of a display

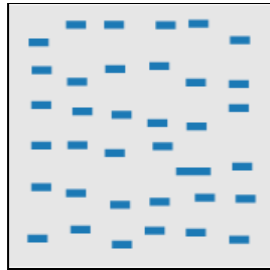


www.csc.ncsu.edu/faculty/healey/PP/index.html

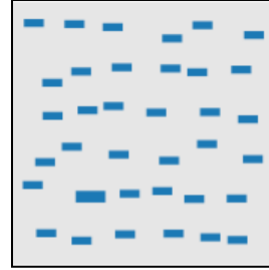
A partial list of pre-attentive visual features



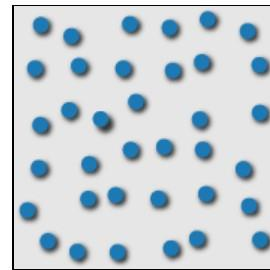
line (blob) orientation
Júlész & Bergen 83; Sagi & Júlész 85a, Wolfe et al. 92; Weigle et al. 2000



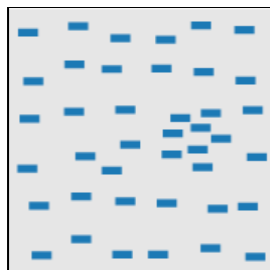
length, width
Sagi & Júlész 85b; Treisman & Gormican 88



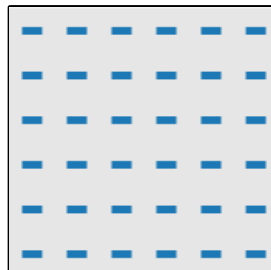
size
Treisman & Gelade 80; Healey & Enns 98; Healey & Enns 99



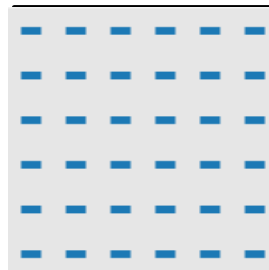
3D depth cues
Enns 90b; Nakayama & Silverman 86



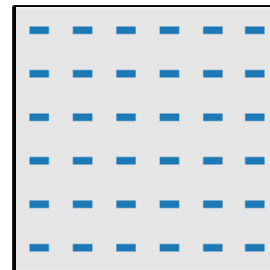
density, contrast
Healey & Enns 98; Healey & Enns 99



velocity of motion
Tynan & Sekuler 82; Nakayama & Silverman 86; Driver & McLeod 92; Hohnsbein & Mateeff 98; Huber & Healey 2005



direction of motion
Nakayama & Silverman 86; Driver & McLeod 92; Huber & Healey 2005



flicker
Gebb et al. 55; Mowbray & Gebhard 55; Brown 65; Júlész 71; Huber & Healey 2005

and many more...

Christopher Healey,

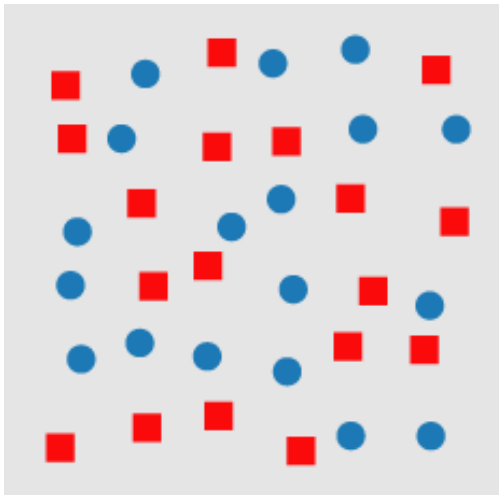
<https://www.csc.ncsu.edu/faculty/healey/PP/index.html>

Multiple Attributes

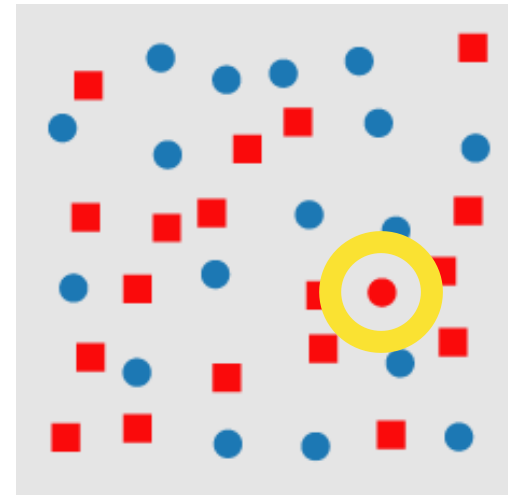
Feature Conjunctions

- A target made up of a combination of non-unique features (a conjunction target) normally cannot be detected pre-attentively
- Where is red circle?

Consistent



Inconsistent

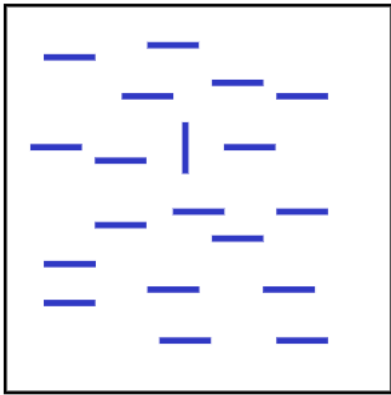


No unique visual property of the target

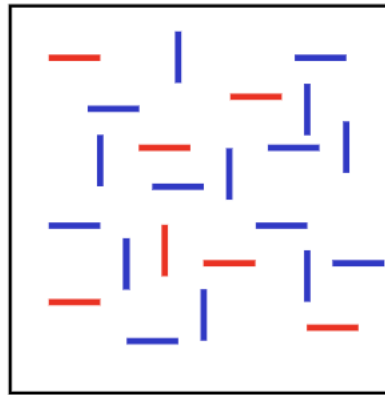
Christopher Healey,

<https://www.csc.ncsu.edu/faculty/healey/PP/index.html>

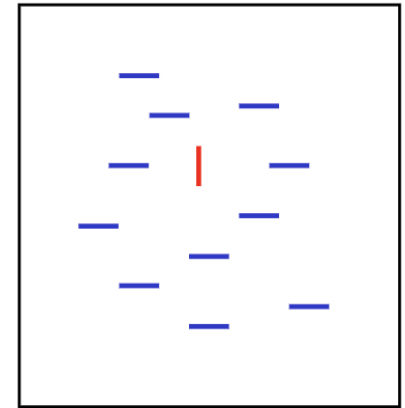
Feature search



Conjunction search



Double feature search



Fast, parallel,
pre-attentive,
effortless, pops
out

Slow, serial,
effortful, needs
attention, does
not pop out

One-Dimensional: Lightness

Classify objects based on lightness



White



White



Black



White



Black

or



White



Black



Black



White



White

One-Dimensional: Shape

Classify objects based on shape



Square



Circle



Circle



Square



Circle

or



Circle



Circle



Square



Circle



Circle

Redundant: Shape & Lightness

Classify objects based on shape. Easier?



Circle



Square



Square

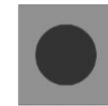


Circle



Square

or



Circle



Square



Square



Square



Circle

Orthogonal: Shape & Lightness

Classify objects based on shape. Difficult?



Circle



Square



Square



Circle



Circle

Orthogonal: Shape & Lightness

Classify objects based on lightness. Difficult?



Circle



Square



Square



Circle

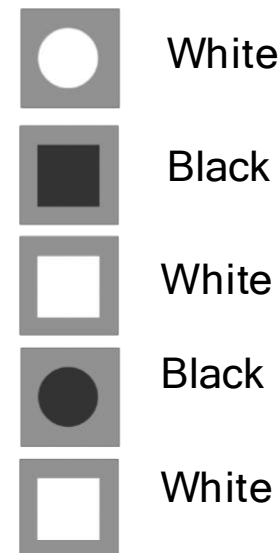
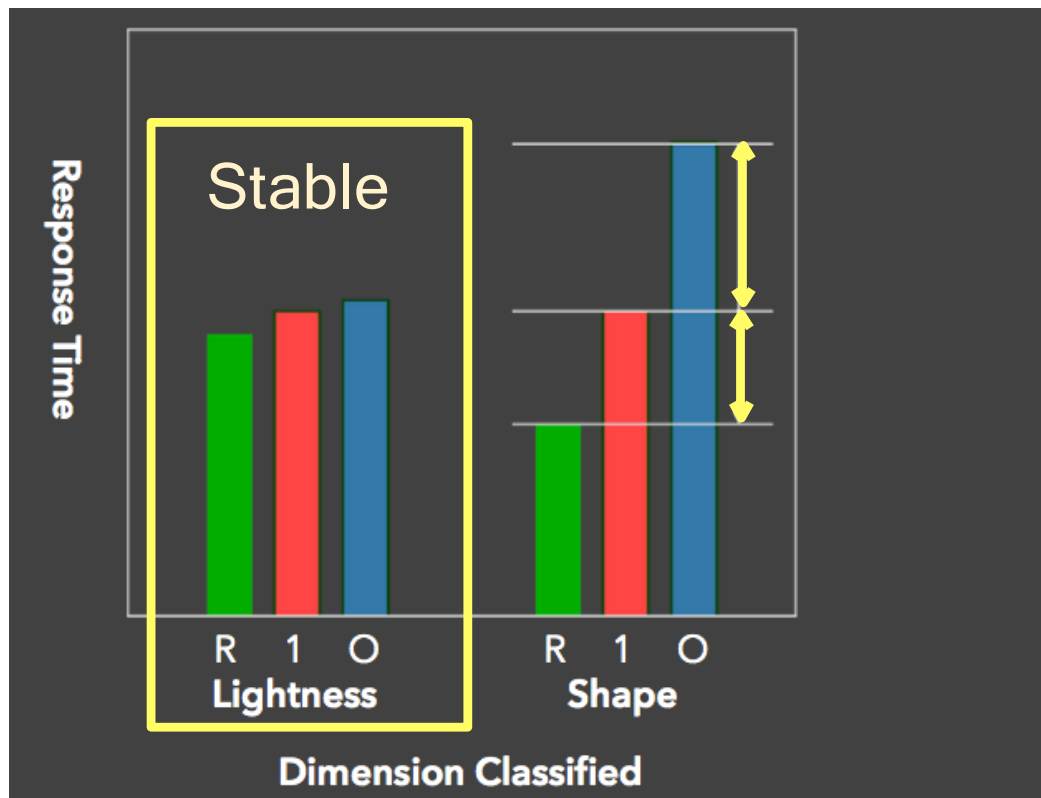


Circle

Speeded Classification

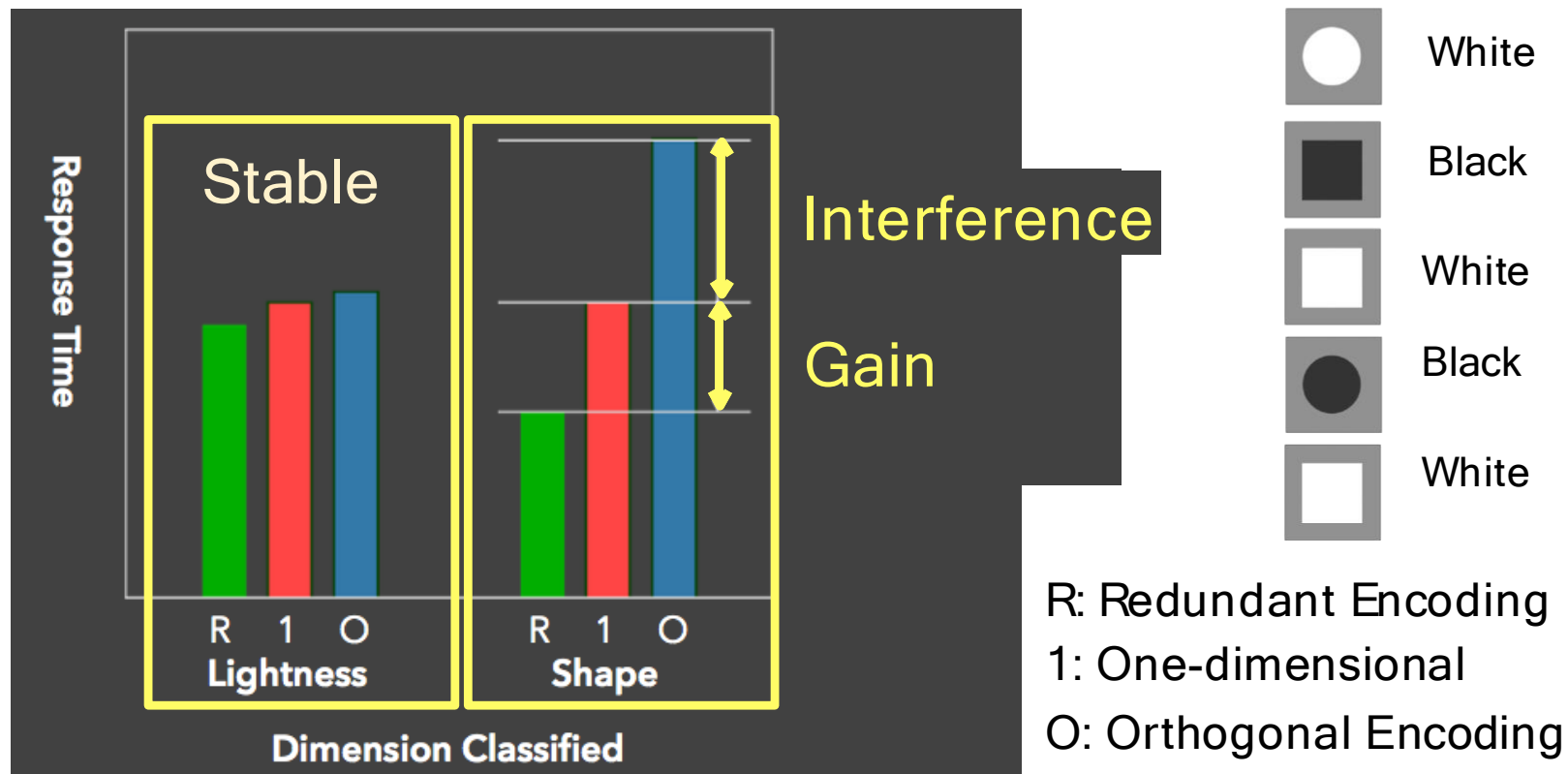
- Redundancy Gain
 - Facilitation in reading one dimension when the other provides redundant information.
- Filtering Interference
 - Difficulty in ignoring one dimension while attending to the other.

Speeded Classification



R: Redundant Encoding
1: One-dimensional
O: Orthogonal Encoding

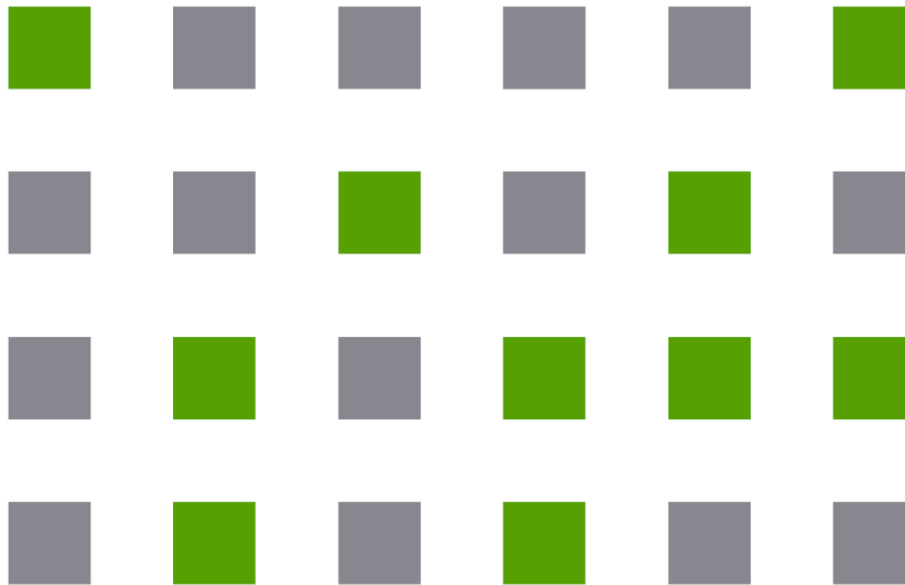
Speeded Classification



Gestalt Grouping

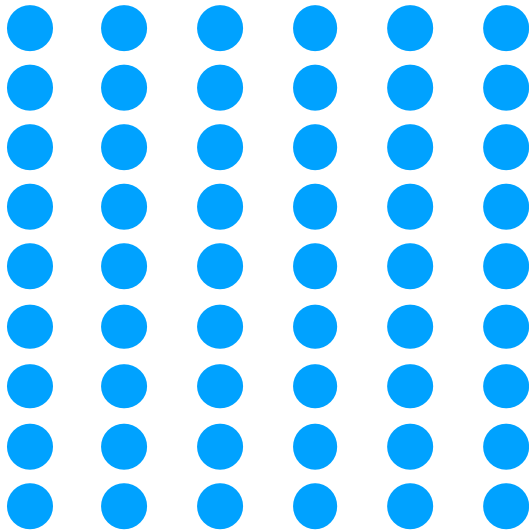
Gestalt principles of design

- Similar elements are visually grouped, regardless of their proximity to each other.
 - They can be grouped by color, shape, or size.

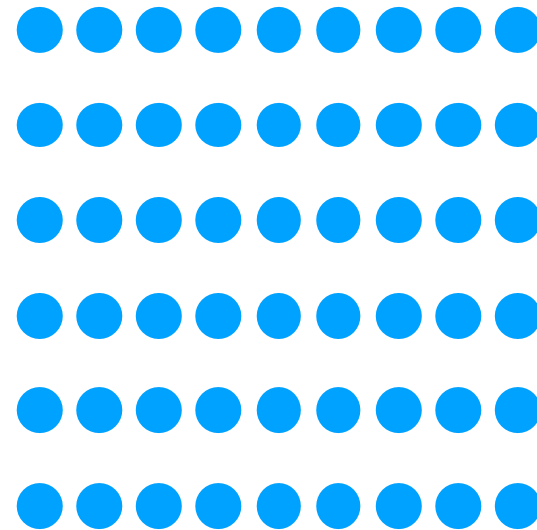


The squares here are all equally spaced and the same size, but we automatically group them by color, even though there's no rhyme or reason to their placement.

Proximity

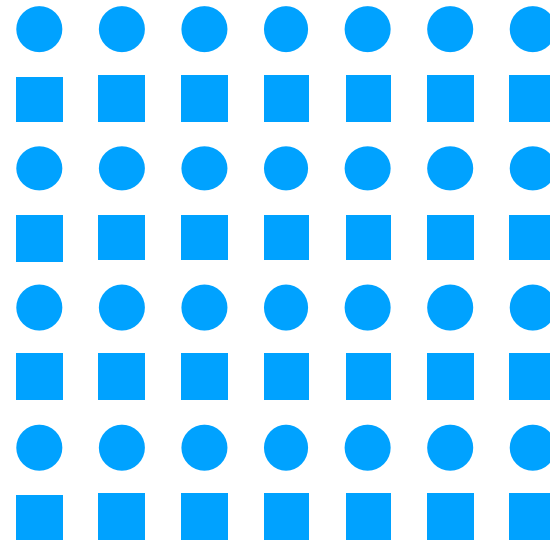
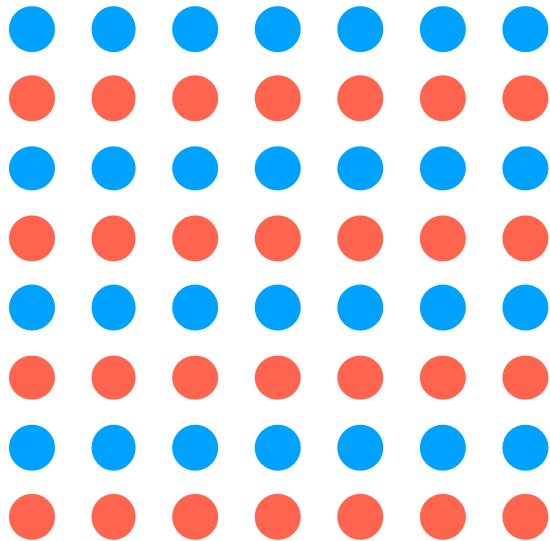


Columns

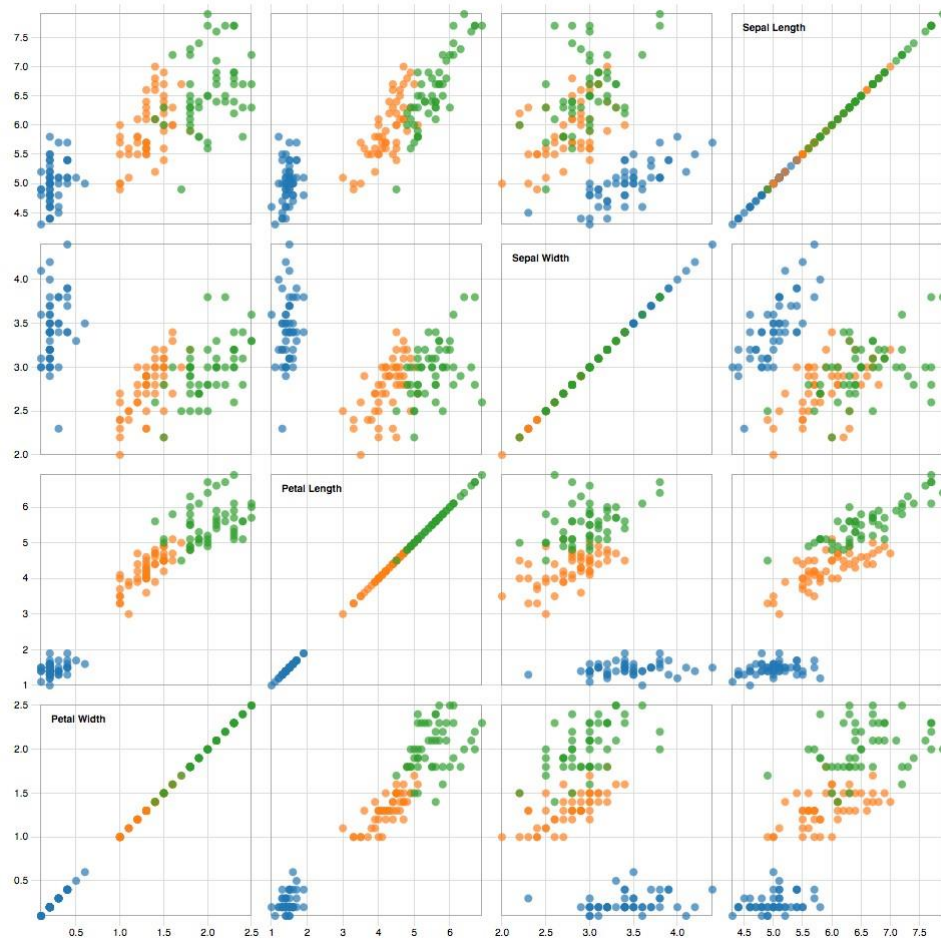


Rows

Similarity



Rows stand out due to similarity.

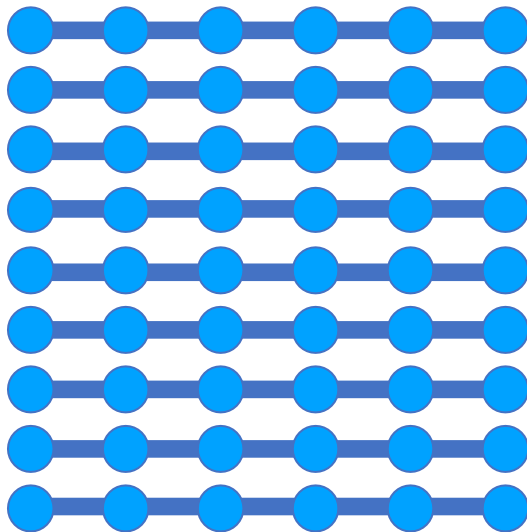


Scatter Plot Matrix

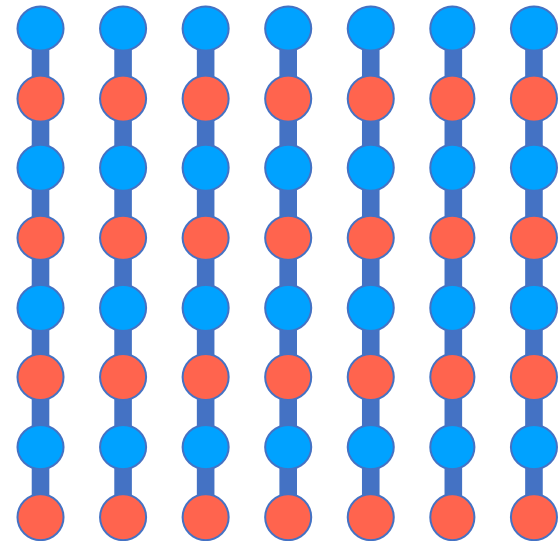
Clusters and outliers

Uniformed Connectedness: Connection

- Connectedness dominates proximity and similarity



Proximity (column)
vs connection (row)



Similarity (row)
vs connection (column)

Uniformed Connectedness: Enclosure

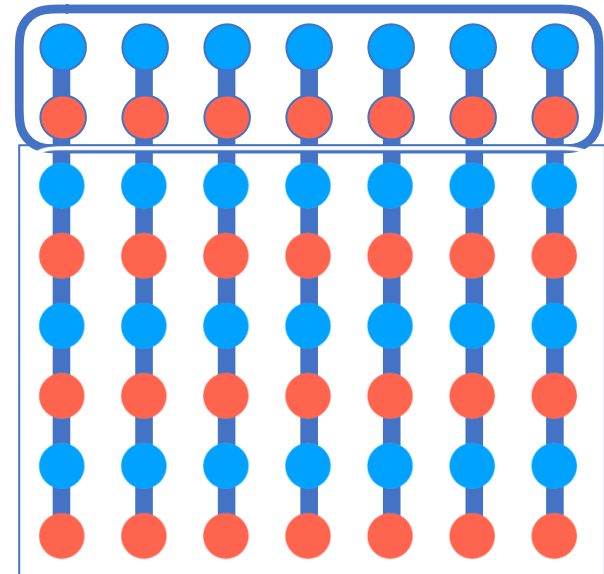
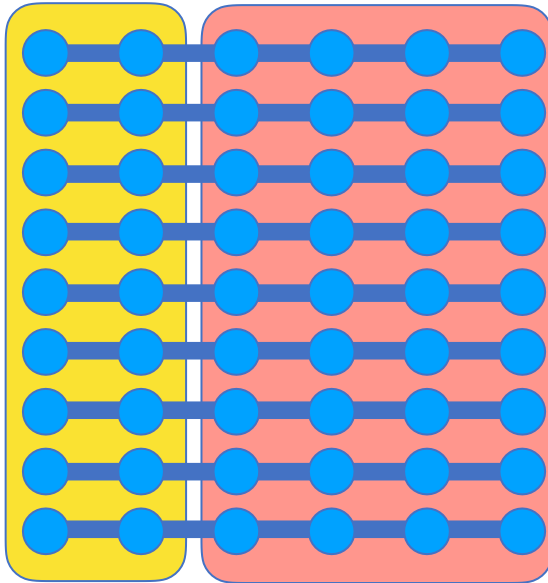
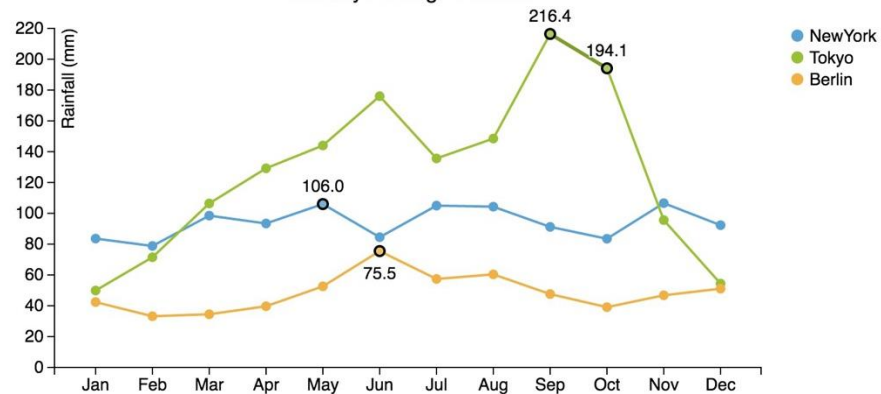
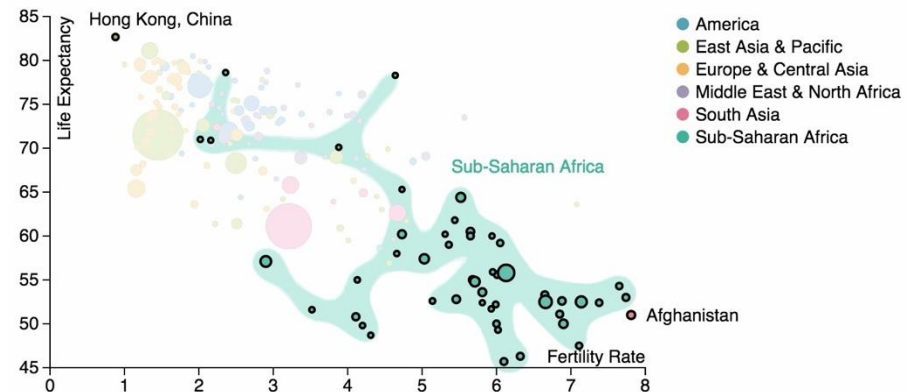


Chart Annotations

Monthly Average Rainfall



Fertility Rate vs. Life Expectancy



Visualizing Sets

- Visualizing sets is a non-trivial problem due to the large number of possible relations between them

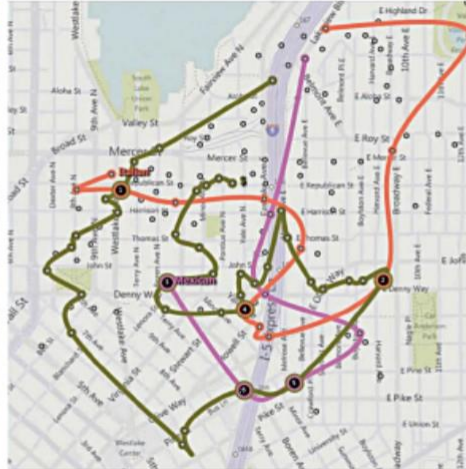
Bubble Sets



Image by [Dinkla et al., 2011]
Technique by [Collins et al., 2009]

https://www.youtube.com/watch?time_continue=34&v=Ju2hSThmPWA&feature=emb_logo

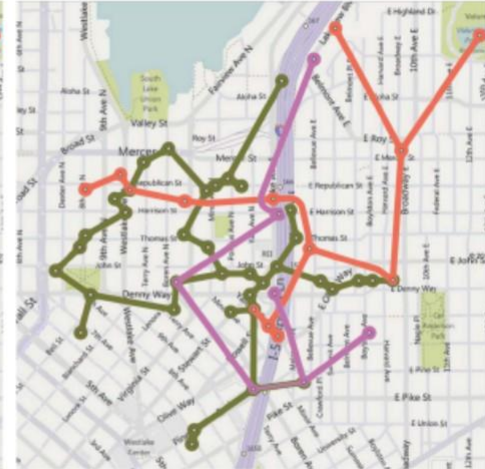
Line Sets



[Alper et al., 2011]

making sure that the traversed path over the elements is relatively short

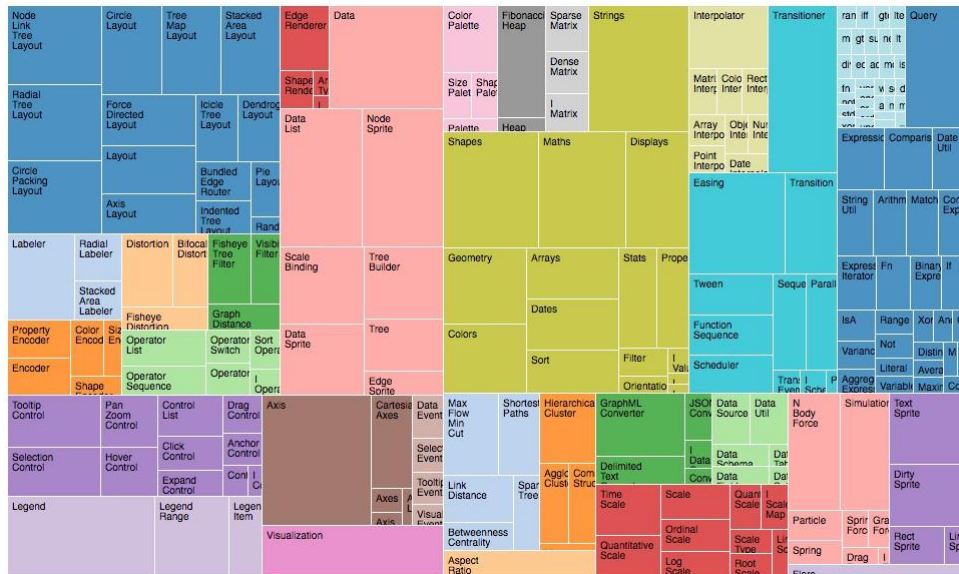
Kelp Diagrams



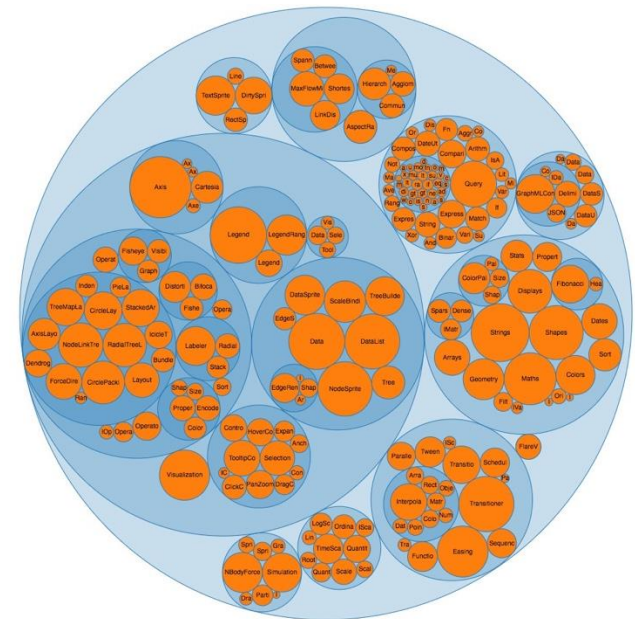
[Dinkla et al., 2012]

[Slides from A. Lex]

TreeMap and Circle Packing



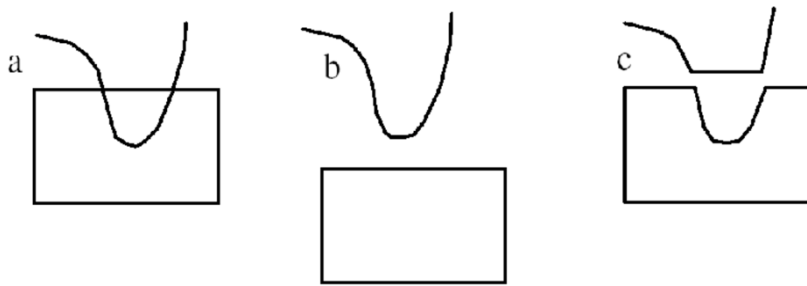
<https://blocks.org/mbostock/4063582>



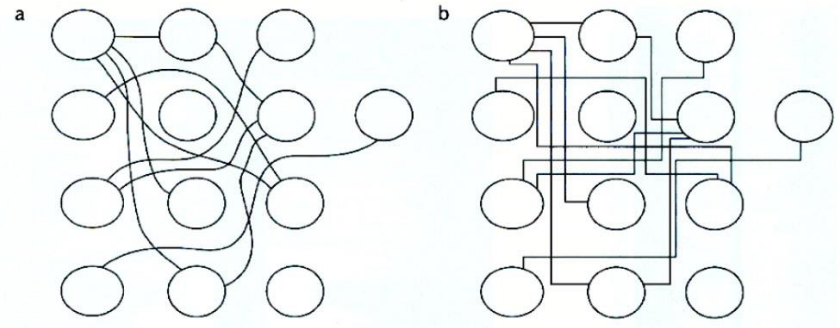
<https://bl.ocks.org/mbostock/4063530>

Proximity, Similarity, Enclosure

Continuity



We prefer smooth not abrupt changes

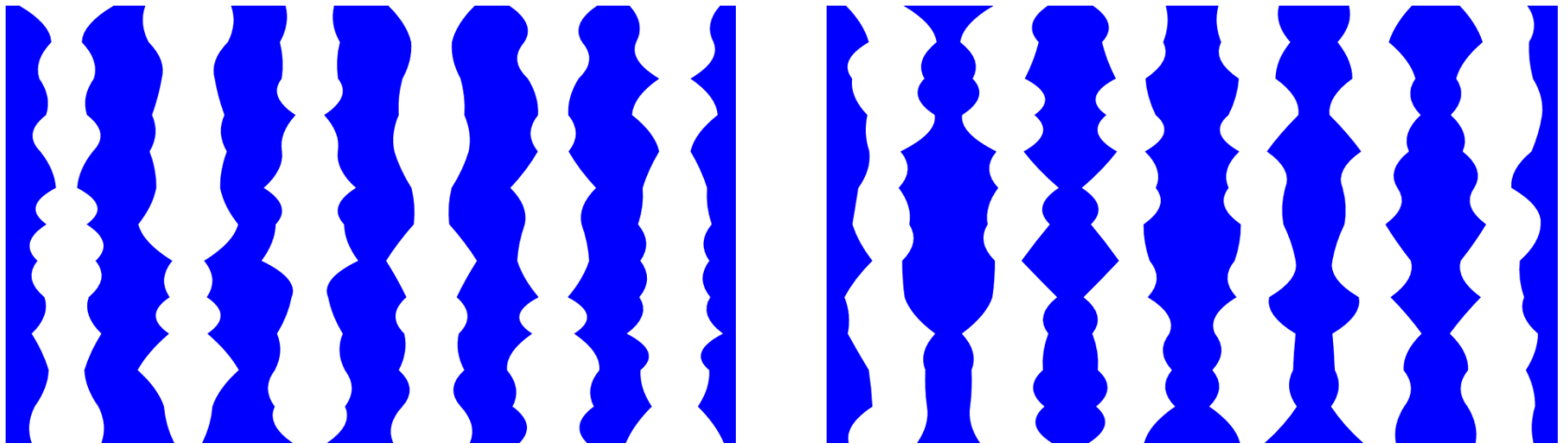


Connections are clearer with smooth contours

[from Ware 04]

Symmetry

- Elements that are symmetrical to each other tend to be grouped together.

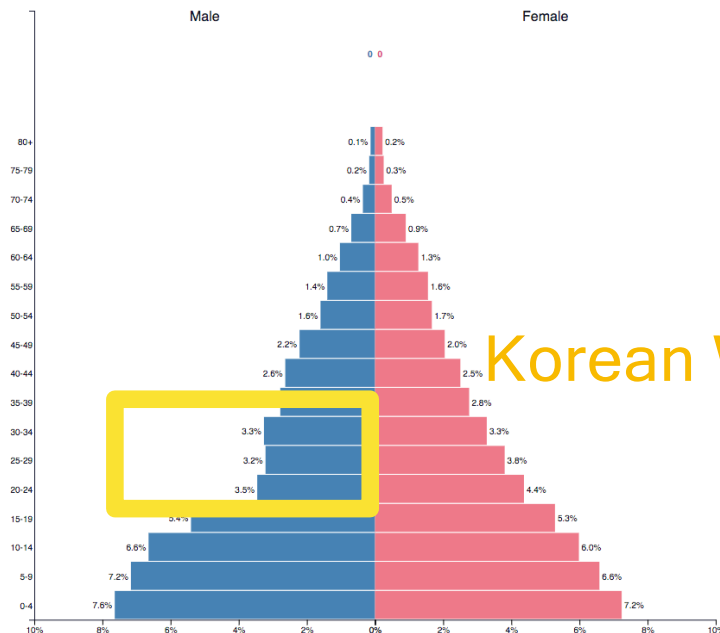


https://isle.hanover.edu/Ch05Object/Ch05SymmetryLaw_evt.html#:~:text=The%20Law%20of%20Symmetry%20is,perceived%20as%20a%20unified%20group.

Population Pyramid (or tornado chart)

Republic of Korea ▼
1953

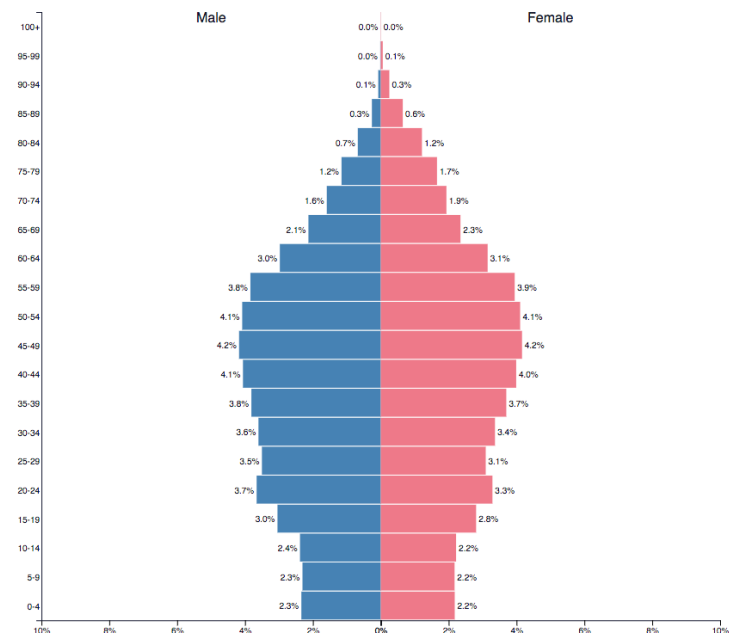
Population: 19,979,069



Korean War?

Republic of Korea ▼
2017

Population: 50,704,971



www.populationpyramid.net/united-states-of-america/2017/

Change Blindness

The phenomenon where even very large changes are not noticed if we are attending to something else.







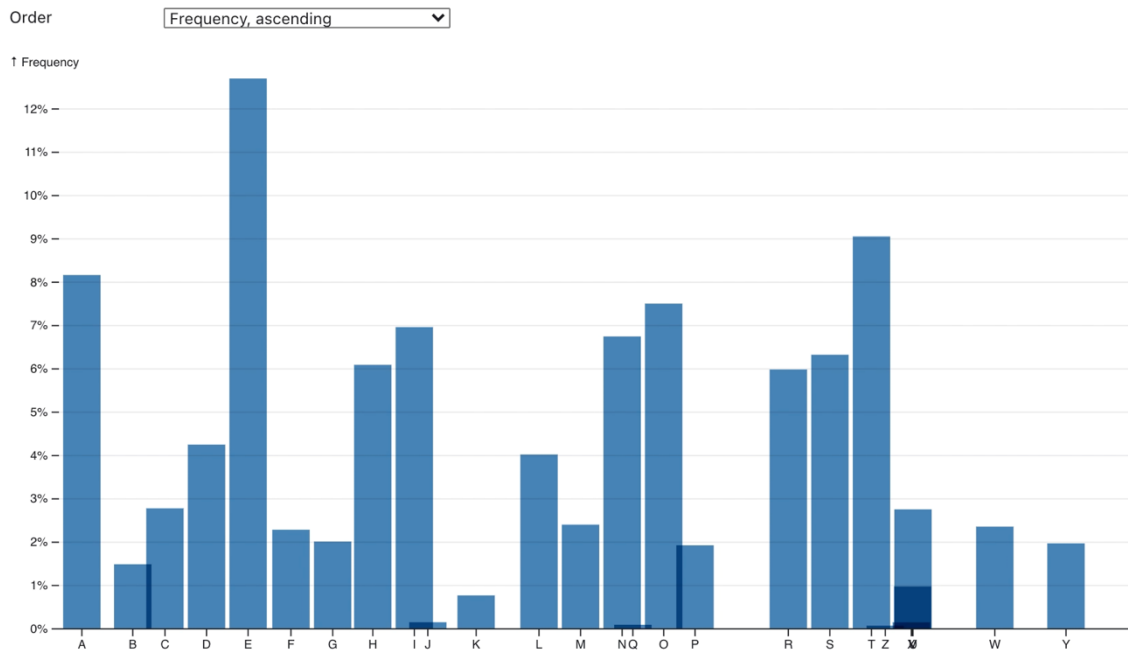


“To see an object change, it is necessary to attend to it.”
– Ronald A. Rensink

Reducing change blindness in visualization

- Provide attentional guidance by leveraging pre-attentive features, Gestalt principles, etc.
- Example: Ease tracking objects through motion

Use the dropdown menu to change the sort order.



<https://bl.ocks.org/mbostock/3885705>

Topics

- Signal Detection
- Magnitude Estimation
- Pre-Attentive Processing
- Using Multiple Visual Encodings
- Gestalt Grouping
- Change Blindness

Take away

- Knowledge of perception can benefit visualization design
 1. Human don't perceive changes and magnitude at face value.
 2. Use pre-attentive visual features for faster target detection.
 3. Be aware of interference and redundancy of multiple features.
 4. Leverage gestalt principles for high-level grouping.
 5. Change blindness in visualization is the failure of design, not because of our vision system.